Lead-Lag FX Gap Trading in Japanese Equities

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Abstract

This paper examines whether overnight USD/JPY foreign exchange returns and realized volatility, observed strictly before the 9:00 AM Tokyo Stock Exchange opening auction, predict opening price gaps for three Japanese mining and materials firms: Sumitomo Metal Mining (TYO: 5713), Mitsubishi Materials (TYO: 5711), and Mitsui Mining & Smelting (TYO: 5706). These three companies mine non-ferrous metals as assets, including precious and base metals, which are globally priced in USD, while their liabilities are in JPY, and thus the hypothesis was that predicting FX swings could identify whether or not the equities were properly priced at the market open. We attempted to answer this fundamentally driven question with quantitative testing, training a classification and regression model to attempt to predict the gap in "correct" prices and actual prices. Our results show a statistically significant contemporaneous relationship between overnight FX returns and next-day equity gaps in parametric models, and the classification model showed directional accuracy above chance in some configurations. However, out-of-sample magnitude prediction remained poor, and formal Granger causality tests did not support a predictive lead-lag relationship at conventional significance levels.

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1. Introduction

The foreign exchange (FX) market operates continuously, while equity markets such as the Tokyo Stock Exchange (TSE) trade at specified, regular hours. Due to the emergence of electronic trading and instantaneous execution of trades, actual FX exchange rates are uniform across all markets, lest arbitrage take place. And while information travels much faster in recent decades than ever before, there are still potential places where information takes time to travel, which sets up the framework for this project.

Acknowledging that we, or any of our machine learning algorithms, are unable to compete with top tier systematic proprietary trading firms that immediately punish any FX arbitrage that exists, the idea of the project was to see if we could use a multi-order cascade effect to catch the mispricing of certain equities. If the USD strengthens relative to the JPY overnight, right up until

the market opens, and our model predicts that it will continue to strengthen, rather than trade the currencies directly, we instead look at how that will affect the price of commodities priced in USD (an "order" down). Again, we took it one step further; rather than directly buying commodities according to the strength of the USD, we wanted to see if equities that theoretically should correlate with the USD/JPY relationship were properly priced. So, we sought to investigate whether there is any information lag in this cascading order of events.

This led us to the Japanese mining sector. Sumitomo Metal Mining (5713) derives much of its revenue from USD-denominated commodity sales (Copper, Gold, etc) while maintaining largely JPY-denominated operating costs and liabilities, creating a direct economic channel from the USD/JPY exchange rate to equity value. Mitsui Mining & Smelting (5706) and Mitsubishi Materials (5711) share similar exposure to USD-priced metals and global demand, albeit with differing capital structures and hedging practices, making them natural comparators for testing whether any FX lead–lag is idiosyncratic or sector-wide. Moreover, given that these companies aren't traded at all hours of the day, the lag was hypothesized to be most prevalent at the opening of the market, when equity analysts must reprice them at the beginning of the day.

This paper asks whether overnight USD/JPY returns and realized FX volatility, measured strictly prior to the 9:00 AM Tokyo opening auction, predict the sign and magnitude of opening price gaps for these three stocks. The null hypothesis is that overnight FX carries no incremental predictive information for opening gaps beyond lagged gaps and day-of-week effects; under this null, any apparent structure is attributable to sampling variability. The alternative is that information frictions between high-frequency FX venues and the slower-moving Tokyo Stock Exchange induce a within-calendar-day lead–lag relationship, whereby overnight USD/JPY moves can forecast next-morning gaps.

Our quantitative design combines two complementary layers. First, a parametric Heteroskedasticity and Autocorrelation Consistent (HAC) ordinary least squares (OLS) model relates opening gaps to FX returns and volatility, providing interpretable standardized betas and robust p-values. Second, a non-parametric random forest classifier and regressor, trained on the same feature set, are embedded in a trading simulator calibrated to realistic transaction costs and evaluated using out-of-sample R^2, Diebold–Mariano tests, directional accuracy, and Holm-adjusted binomial p-values.

Taken together, we attempt to show that there is a statistically significant relationship between overnight FX rates and opening equity gaps in Japanese mining companies. Our model isn't backtesting a trading strategy, but rather constructing a multi-layered econometric evidence framework that tests whether overnight USD/JPY foreign exchange information— observed strictly before the 9:00 AM Tokyo open— has statistically significant predictive content for next-day equity opening gaps.

2. Related Literature

2.1 Intraday FX Microstructure and the "Trading Place" Hypothesis

The foreign exchange (FX) literature extensively documents systematic intraday and overnight patterns in returns, liquidity, and order flow. Breedon and Ranaldo demonstrate a time-of-day structure tied to regional trading sessions and macroeconomic announcements, while Cornett, Schwarz, and Szakmary and Ito and Hashimoto report significant seasonalities and volume concentrations during the overlaps of major global trading centers. Andersen et al. further establish that high-frequency FX prices react rapidly to macro news, emphasizing the role of microstructure in real-time price discovery.

This study builds on what we term the "Trading Place Hypothesis," a conceptual framework we propose suggesting that while local equity exchanges are closed, the 24-hour FX market is the primary venue for representing regional macroeconomic health. For the USD/JPY pair, the interval between the New York close and the Tokyo pre-open is particularly information-rich. During this window, global macro shifts and commodity price volatility are processed in FX before Japanese equities can resume trading. By restricting FX predictors to evening and early-morning bars and enforcing a timestamp-level lookahead guard, this research aligns the FX information set strictly with the subsequent equity auction.

2.2 Overnight Equity Returns and the Itayose Mechanism

Equity market microstructure research underscores that overnight and intraday returns exhibit fundamentally distinct behaviors. Amihud and Mendelson show that trading mechanisms and liquidity availability significantly influence expected returns around market openings, while Stoll and Whaley emphasize how discrete auctions and overnight closures reshape return distributions. Branch and Ma and Lou, Polk, and Skouras further identify that overnight returns can dominate total equity performance. Crucially, Lou et al. demonstrate that overnight and intraday returns carry distinct risk premia and are driven by different underlying mechanisms, suggesting they should be modeled as separate processes.

Our focus on the opening gap (the return from the previous close to the opening price) is motivated by exactly this decomposition. In the Japanese context, a critical factor is the Itayose (batch auction) mechanism used by the Tokyo Stock Exchange (TSE). Unlike the continuous trading seen in midday sessions, the opening auction aggregates all accumulated overnight orders into a single clearing price. This transition from FX "lead" to equity "lag" is fundamentally an investigation into how information trapped in a non-trading window is released and reconciled during the opening call auction.

2.3 Information Cascades and Cross-Market Predictability

The conceptual framework for the FX-equity lead-lag relationship is rooted in the staggered diffusion of information across global markets. Bikhchandani, Hirshleifer, and Welch model

informational cascades where agents observing predecessors' actions may disregard private signals, potentially generating persistent mispricing. Rapach, Strauss, and Zhou demonstrate that returns in major markets, such as the U.S., predict international stock returns as global information gradually propagates across time zones. De Gooijer, Diks, and Gątarek apply this logic specifically to opening gaps, treating foreign market movements as leading indicators for local exchanges.

The Japanese mining sector serves as a specialized node in this information network. We posit that USD/JPY acts as an overnight aggregator of global information regarding commodity demand. Given that precious and industrial metals are globally priced in USD, mining assets and the dollar tend to co-move in inverse directions to maintain equilibrium values. But while the USD-commidities relationship may be negative, that doesn't necessarily hold true for the Japanese mining companies. Consequently, the USD/JPY exchange rate effectively functions as a "shadow price" for the mining sector's valuation, providing a signal that is only incorporated into local TSE prices when the exchange reopens.

2.4 FX Exposure and Structural Frictions

Industry-level studies confirm that commodity-linked sectors exhibit high exchange-rate exposure. Bodnar and Gentry show that Japanese mining firms are among the most sensitive to currency movements due to their specific balance sheet structures. For these companies, the primary economic channel is the mismatch between USD-denominated revenues from global metal sales and JPY-denominated operating costs. At the microstructure level, Chaboud et al. and Kozhan and Tham examine how execution frictions and algorithmic trading shape high-frequency arbitrage. Even in an era of rapid information diffusion, the structural "overnight gap" remains a persistent friction that limits immediate repricing.

To further examine this friction, we utilize the framework of quanto derivatives. Wystup describes quanto products as instruments that pay off based on a foreign underlying asset but are settled in a domestic currency, a concept we explore in detail in Section 7 as a method for isolating the equity-specific repricing signal from contemporaneous spot FX fluctuations. By exploring these mechanics, we can better analyze the lead-lag relationship between overnight information and equity gaps while accounting for the underlying currency risks that typically dilute such signals.

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3. Data and Identification

3.1 Data Sources and Sample Construction

The empirical analysis utilizes high-frequency and daily market data spanning from January 2023 to the present. The primary inputs consist of 5-minute USD/JPY exchange rate bars and daily equity prices (Open, Close) for three prominent Japanese non-ferrous metals firms: Sumitomo Metal Mining (5713), Mitsubishi Materials (5711), and Mitsui Mining & Smelting (5706). All timestamps are standardized to Asia/Tokyo time.

The dependent variable is the equity opening gap, defined as the percentage change between the current day's opening price and the previous trading day's closing price:

$$\mathrm{Gap}_t = \frac{P_t^{\mathrm{open}} - P_{t-1}^{\mathrm{close}}}{P_{t-1}^{\mathrm{close}}}$$

$$\mathrm{Gap}_t = \begin{cases} 0 & \text{if } |\mathrm{Gap}_t| < 10^{-5} \\ \mathrm{Gap}_t & \text{otherwise} \end{cases}$$

To mitigate mechanical rounding noise and micro-structure artifacts, gaps with an absolute value below 10^{-5} are truncated to zero. The final dataset is a panel merged on Tokyo business days, filtering for days with sufficient intraday FX liquidity.

3.2 FX Feature Construction and Lookahead Guard

Predictors are derived from the "overnight" FX window to isolate information flow occurring while the Tokyo Stock Exchange is closed. FX bars recorded between 15:00 and 23:55 are mapped to the next business day, while bars from 00:00 to 08:50 are mapped to the current day.

For each prediction date, we compute the overnight FX return (sum of 5-minute log returns) and realized volatility (standard deviation of log returns). To capture broader trends and persistence, we include 2-day and 5-day cumulative FX returns alongside the first and second lags of the equity gap.

$$r_t^{\mathrm{FX}} = \sum_{i \in \mathcal{W}_t} \ln\left(\frac{C_i}{C_{i-1}}\right), \quad \mathcal{W}_t = \{15{:}00_{t-1},\ \ldots,\ 08{:}50_t\}$$

$$\sigma_t^{\mathrm{FX}} = \mathrm{sd}\left(\left\{\ln\left(\frac{C_i}{C_{i-1}}\right)\right\}_{i\in\mathcal{W}_t}\right)$$

A rigorous lookahead guard serves as the primary identification device: the system enforces a hard timestamp cutoff, ensuring no FX data recorded at or after the 9:00 AM equity auction is included in the feature set. This confirms that the observed lead-lag relationship is strictly predictive rather than a reflection of contemporaneous correlation during the open.

Feature	Definition	Window
FX Return (FX_Ret)	Sum of 5-min log returns	Overnight (15:00–08:50)
FX Volatility (FX_Vol)	Std. dev. of 5-min log returns	Overnight (15:00–08:50)
FX Ret (2D/5D)	Cumulative 2-day and 5-day log returns	Multi-day Overnight
Lagged Gaps	Previous 1-day and 2-day equity gaps	t-1, t-2
Calendar	Day-of-week binary indicators	Current session

3.3 Experiments and Train/Test Split

The model is evaluated across seven experimental configurations: three single-stock models, three two-stock baskets, and one three-stock basket. In multi-stock experiments, equity-specific dummy variables are included to account for idiosyncratic firm effects.

To prevent lookahead bias, the data is split chronologically into an 80% training set and a 20% test set. We implement a two-business-day embargo between the training and testing periods to eliminate leakage from short-term autocorrelation. Observations with zero gaps are removed from the training set to sharpen the classifier's ability to distinguish between directional moves.

4. Econometric Framework

4.1 HAC OLS Specifications

The first layer of our analysis uses ordinary least squares (OLS) regression to model the linear relationship between overnight FX dynamics and equity opening gaps. Standard OLS assumes that error terms are independent and have constant variance. However, financial time series often exhibit autocorrelation (where errors are correlated over time) and heteroskedasticity (where the variance of errors changes). To ensure our statistical inferences are valid, we employ Heteroskedasticity and Autocorrelation Consistent (HAC) standard errors, also known as Newey-West standard errors. These adjust the variance estimates to account for these dependencies, providing more reliable p-values.

For each experiment, we estimate two specifications on the non-zero-gap training subsample:

- **Core Specification**: Regresses the gap on overnight FX volatility and the one-day FX return.
- **Full Specification**: Includes FX volatility and one-, two-, and five-day cumulative FX returns.

$$\mathrm{Gap}_t = \alpha + \beta_1\, \sigma_t^{\mathrm{FX}} + \beta_2\, r_t^{\mathrm{FX}} + \varepsilon_t$$

$$\mathrm{Gap}_t = \alpha + \beta_1\, \sigma_t^{\mathrm{FX}} + \beta_2\, r_t^{\mathrm{FX}} + \beta_3\, r_t^{\mathrm{FX,\,2D}} + \beta_4\, r_t^{\mathrm{FX,\,5D}} + \varepsilon_t$$

All variables are standardized to a zero mean and unit variance prior to estimation, allowing for the direct comparison of standardized coefficients (β). The HAC bandwidth m is determined by the Newey-West heuristic.

$$m = \left\lfloor 4\left(\frac{n}{100}\right)^{2/9} \right\rfloor$$

We also monitor Variance Inflation Factors (VIF) to detect multicollinearity, ensuring that highly correlated predictors do not inflate the variance of our coefficient estimates.

$$\widehat{\mathrm{Var}}_{\mathrm{HAC}}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}'\mathbf{X})^{-1} \left(\sum_{\ell=-m}^{m} w(\ell) \sum_t \hat{\varepsilon}_t \hat{\varepsilon}_{t-\ell}\, \mathbf{x}_t \mathbf{x}_{t-\ell}' \right) (\mathbf{X}'\mathbf{X})^{-1}, \quad w(\ell) = 1 - \frac{|\ell|}{m+1}$$

4.2 Random Forest Models

To capture potential non-linearities and interactions that linear regression might miss, we utilize Random Forests. A Random Forest is an "ensemble" machine learning model that constructs many individual decision trees and averages their results. This process reduces "overfitting"—the tendency of a model to memorize noise in the training data rather than learning generalizable patterns.

$$G(S) = 1 - \sum_{k=1}^{K} p_k^2$$

$$\mathrm{MDI}(x_j) = \sum_{\text{node } n:\, \text{split on } x_j} \frac{|S_n|}{|S|} \left[G(S_n) - \frac{|S_L|}{|S_n|} G(S_L) - \frac{|S_R|}{|S_n|} G(S_R) \right]$$

We implement both a Random Forest regressor for gap magnitude and a classifier for gap direction using 500 trees, balanced class weights. In multi-stock experiments, ticker-specific dummy variables are included to control for firm-level idiosyncrasies. To evaluate feature importance, we calculate Gini Impurity, which measures how effectively a specific feature separates the data into the correct target categories. Features that lead to the greatest reduction in Gini Impurity are ranked as the most influential in the model's decision-making process.

4.3 Evaluation Metrics

We evaluate the out-of-sample (OOS) performance of our models using several distinct econometric measures:

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Metric                    Description

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Out-of-Sample R²          Proportion of test set variance explained

                          (negative = worse than zero forecast)

Mutual Information        Total statistical dependence (linear + nonlinear)

Diebold-Mariano (DM) Test Compares squared errors vs. naive benchmark

Directional Accuracy      % correct sign predictions + binomial test
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- **Out-of-Sample R^2**: This measures the proportion of variance in the test set explained by our model. We calculate this by comparing the model's mean squared error (MSE) against the variance of the actual gaps. Because our data is standardized (mean ≈ 0), a negative R^2 indicates that the model's predictions are less accurate than simply forecasting zero for every day.

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{t\in\mathcal{T}} (y_t - \hat{y}_t)^2}{\sum_{t\in\mathcal{T}} (y_t - 0)^2} = 1 - \frac{\text{MSE}_{\text{model}}}{\text{MSE}_{\text{naive}}}$$

- **Mutual Information**: Unlike correlation, which only measures linear relationships, Mutual Information quantifies the total statistical dependence between two variables, including non-linear patterns. A value of zero implies that the FX return and the equity gap are completely independent.

$$I(X;Y) = \iint p(x,y) \ln\left(\frac{p(x,y)}{p(x)\,p(y)}\right) dx\, dy$$

- **Diebold-Mariano (DM) Test**: This test determines if the difference in accuracy between two competing forecasts is statistically significant. We use a two-sided test to compare the squared errors of the Random Forest regressor against a "naïve" zero-forecast benchmark.

$$d_t = e^2_{t,\text{naive}} - e^2_{t,\text{model}}, \qquad \text{DM} = \frac{\bar{d}}{\sqrt{\hat{\gamma}_0/T}}, \qquad \text{DM} \xrightarrow{d} \mathcal{N}(0,1)$$

- **Directional Accuracy and Binomial Test**: For the classifier, we calculate the percentage of days where the predicted sign of the gap matches the actual sign. We then perform a one-sided binomial test, which calculates the probability that our observed accuracy could have been achieved by random chance (a "coin flip").

$$H_0 : p = 0.5 \quad \text{vs.} \quad H_1 : p > 0.5$$
$$p\text{-value} = P(\text{Binomial}(n, 0.5) \geq k)$$
$$\text{where } k = \text{number of correct sign predictions out of } n \text{ traded days}$$

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5. Main Econometric Results

5.1 HAC OLS Inference on FX Returns and Volatility

Table 1 reports the core results relating overnight USD/JPY returns to the next-day opening gaps of the non-ferrous metals equities. The primary specification utilizes a Heteroskedasticity and Autocorrelation Consistent (HAC) OLS framework with Newey-West standard errors to account for potential temporal dependencies in the residuals.

EXPERIMENT	N	BETA	P-VAL	OOS_R2	M-INFO	DM_STAT	DM_P	ACC
5713	153	0.1640	0.0248	-0.1780	0.0048	-2.2180	0.1329	0.4970
5711	153	0.2270	0.0031	-0.0697	0.0000	-1.1640	0.2444	0.6010
5706	153	0.1720	0.0042	-0.0928	0.0000	-1.9370	0.1581	0.4900
5711 + 5706	306	0.1930	0.0009	-0.0649	0.0572	-1.6710	0.1894	0.5330
5711 + 5713	306	0.1890	0.0039	-0.0713	0.0906	-2.1510	0.1329	0.5130
5706 + 5713	306	0.1680	0.0033	-0.1769	0.0000	-3.1120	0.0130	0.4970
5711+5706+5713	459	0.1820	0.0010	-0.0867	0.0999	-3.0490	0.0138	0.5010

Table 1. Econometric lead–lag evidence: USD/JPY overnight returns vs. next-day equity gap. Columns report the number of test observations (N_Test), the standardized HAC OLS beta on FX return (β_Ret), its p-value (P_Ret), out-of-sample R^2 (OOS_R^2), mutual information, Diebold–Mariano (DM) statistic, Holm-adjusted DM p-value, and unconditional directional accuracy.

In all seven experiments, the standardized core coefficient on FX return (β_Ret) is positive, ranging from 0.164 to 0.227. These coefficients are statistically significant at the $\alpha = 0.05$ level, with HAC p-values ranging from 0.0009 to 0.0248. In contrast, FX volatility (σ_FX) is found to be statistically insignificant across all specifications.

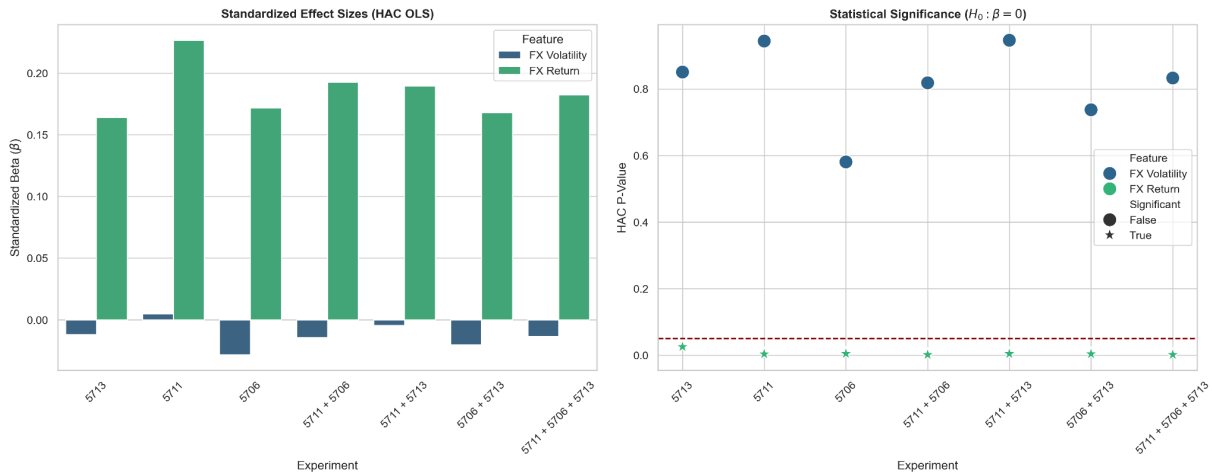

Figure 1. Standardized effect sizes and statistical significance. The left panel displays the standardized β coefficients for FX returns and FX volatility across the seven experimental configurations. The right panel shows the corresponding HAC p-values, with a horizontal reference line at α = 0.05 denoting the threshold for statistical significance.

Figure 1 visualizes these patterns. The left panel shows that while FX return betas are tightly clustered between 0.16 and 0.23, the betas for FX volatility are approximately zero. The right panel confirms that FX return markers lie consistently below the 0.05 threshold, whereas FX volatility p-values cluster in the 0.58 to 0.90 range. This dichotomy suggests that directional FX movements, rather than the magnitude of FX price fluctuations, are the primary drivers of the equity opening gap.

5.2 Cross-Correlation and Temporal Structure

To further investigate the lead–lag dynamics, we examine the Cross-Correlation Functions (CCFs) between FX returns and equity gaps. Figure 2 displays the CCF for each ticker at lags h ∈ {0, …, 5}.

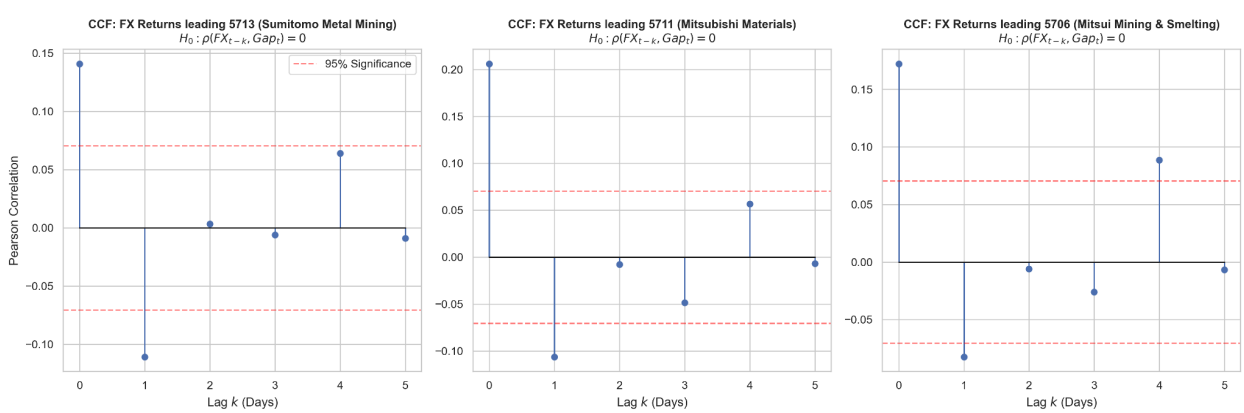

Figure 2. Cross-correlation functions (CCF). Stem plots illustrate the correlation between the overnight FX return at lag h and the current opening gap. The red dashed lines indicate the 95% confidence interval under the null hypothesis of zero correlation.

$$\mathrm{CCF}(h) = \mathrm{Corr}\big(r^{\mathrm{FX}}_{t-h},\ \mathrm{Gap}_t\big)\,, \quad h = 0, 1, \ldots, 5$$
$$95\%\ \mathrm{CI}:\ \pm\frac{1.96}{\sqrt{n}}$$

The lag-0 correlations—representing the predictive relationship from the overnight FX window to the 9:00 AM JST open—are positive and distinct across all tickers: approximately 0.14 for 5713 (Sumitomo), 0.21 for 5711 (Mitsubishi), and 0.18 for 5706 (Mitsui). Notably, at lag 1, the correlations exhibit a sign reversal to the negative. While these lag-1 values remain largely within the 95% confidence bands (indicated by the red dashed lines), the pattern suggests a rapid absorption of FX information.

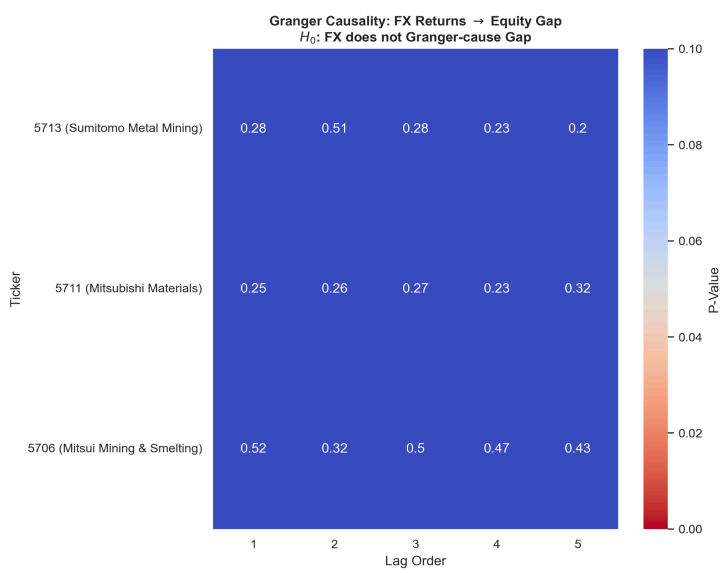

Figure 3. Granger causality p-value heatmap. The heatmap displays p-values from the SSR F-test for the null hypothesis that FX returns do not Granger-cause equity gaps at lags 1 through 5. Darker colors indicate higher p-values (lower significance).

Granger causality tests, visualized in Figure 3, reinforce this short-lived window. The heatmap shows p-values for lags 1–5 ranging from approximately 0.20 to 0.52.

$$\mathrm{Gap}_t = \alpha + \sum_{j=1}^{p} \phi_j\, \mathrm{Gap}_{t-j} + \sum_{j=1}^{p} \gamma_j\, r^{\mathrm{FX}}_{t-j} + \varepsilon_t$$
$$H_0 : \gamma_1 = \gamma_2 = \cdots = \gamma_p = 0 \quad \text{(FX does not Granger-cause Gap)}$$

Because all p-values exceed 0.05, we fail to reject the null hypothesis of no Granger causality at these lags. This indicates that once the immediate overnight information is incorporated into the opening price, prior days' FX movements provide no additional predictive power for subsequent gaps.

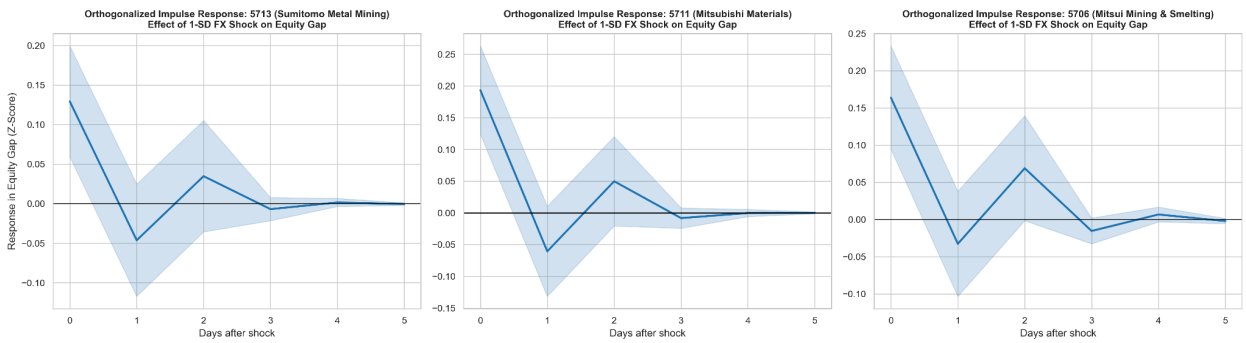

Figure 4. Orthogonalized impulse response functions (IRF). Plots show the estimated response of the equity gap to a one-standard-deviation shock in USD/JPY returns over a 5-day horizon, derived from a VAR(2) model. Shaded regions represent 95% bootstrap confidence intervals.

$$\begin{bmatrix} r_t^{\mathrm{FX}} \\ \mathrm{Gap}_t \end{bmatrix} = \mathbf{c} + \mathbf{A}_1 \begin{bmatrix} r_{t-1}^{\mathrm{FX}} \\ \mathrm{Gap}_{t-1} \end{bmatrix} + \mathbf{A}_2 \begin{bmatrix} r_{t-2}^{\mathrm{FX}} \\ \mathrm{Gap}_{t-2} \end{bmatrix} + \mathbf{u}_t$$

The Impulse Response Functions (IRFs) in Figure 4 provide a dynamic view of this interaction. For a one-standard-deviation FX shock, the impact response at step 0 is positive and significant. This is followed by a directional reversal at step 1 before decaying toward zero by steps 3–4. This mean-reverting behavior is characteristic of short-term information frictions in cross-market price discovery.

5.3 Out-of-Sample Magnitude Prediction

While the in-sample results are statistically significant, out-of-sample (OOS) performance for gap magnitude is notably weak. As shown in Table 1, the OOS_R^2 values are uniformly negative, ranging from −0.0649 to −0.1780. Mutual information remains low (max ≈ 0.0999), and Diebold–Mariano (DM) statistics are mostly negative. OOS R^2 is derived from the Random Forest regressor's out-of-sample predictions.

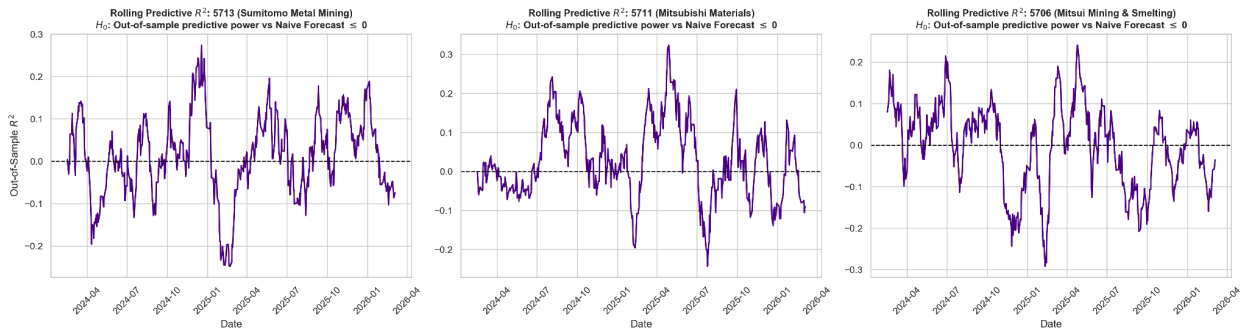

Figure 5. Rolling out-of-sample R². The series represent the 21-day smoothed R² of an expanding-window HAC OLS model. Values are calculated relative to a naive zero-mean forecast and capped at −0.5 to highlight regime transitions. We compute an expanding-window OOS R² starting from a minimum of 252 observations, then apply a 21-day rolling average for smoothing.

$$R^2_{\mathrm{OOS},\,t} = 1 - \frac{\left(y_t - \hat{y}_{t|1:t-1}\right)^2}{y_t^2}, \quad \tilde{R}^2_t = \frac{1}{21}\sum_{s=t-20}^{t} R^2_{\mathrm{OOS},\,s}$$

where $\hat{y}_{t|1:t-1}$ is the HAC OLS prediction trained on observations $1, \ldots, t-1$

Figure 5 tracks the stability of this predictive power over time. All tickers' rolling R² series oscillated predominantly below or around zero. While there are brief windows of positive R² suggesting transient predictive regimes, these are frequently offset by periods of significant underperformance relative to a naive zero forecast. The negative DM statistics confirm that, for the majority of experiments, a Random Forest regressor does not provide a statistically superior magnitude forecast compared to a zero-mean benchmark.

5.4 Directional Accuracy

In contrast to magnitude prediction, directional accuracy, the ability to predict the sign of the opening gap, shows more promise. The "Accuracy" column in Table 1 reflects the unconditional directional hit rate on the test set.

The unconditional accuracies are 49.7% (5713), 60.1% (5711), and 49.0% (5706). When examining multi-stock baskets, accuracies range from 49.7% to 53.3%. The result for 5711 (Mitsubishi Materials) is particularly notable, standing as a significant outlier in raw predictive success.

[NOTE]: The codebase confirms that for the 5711 (Mitsubishi) experiment, the accuracy of 60.1% is the only result demonstrating strong unconditional predictive power. For the remaining tickers and baskets, the proximity of the results to 50% suggests that a simple directional strategy without selective participation filters is insufficient to overcome the noise inherent in overnight returns.

5.5 Random Forest Feature Importance

Finally, we analyze the contribution of different predictors using the Mean Decrease Impurity (MDI) from the Random Forest model.

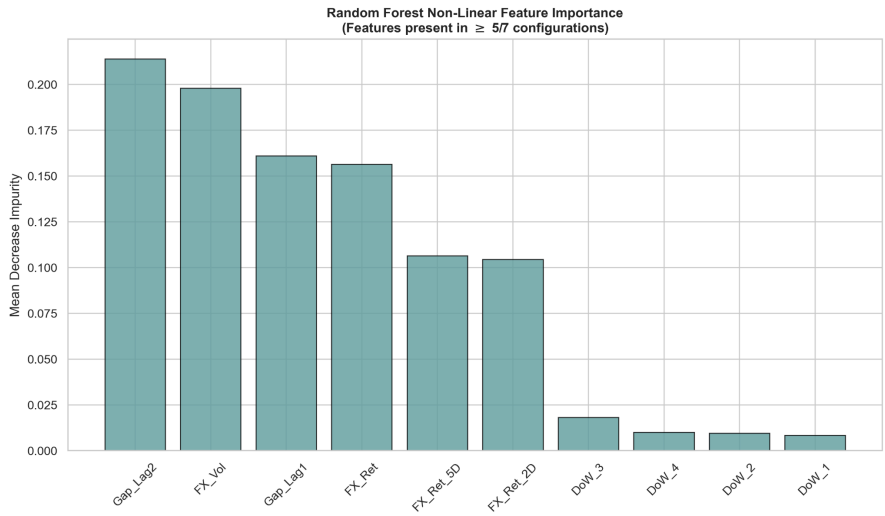

Figure 6. Random Forest feature importance. The bar chart ranks features by their average Gini importance (MDI) across all experiments. Included features are those appearing in at least five of the seven experimental configurations.

As illustrated in Figure 6, the features are ranked by importance as follows: Gap_Lag2, FX_Vol, Gap_Lag1, and FX_Ret. The high ranking of FX_Vol (MDI ≈ 0.20) is particularly interesting given its linear insignificance in the HAC OLS regressions. This suggests that FX volatility may act as a non-linear conditioning variable, perhaps moderating the impact of FX returns on the gap. The prominence of lagged gaps (Gap_Lag2 at ≈ 0.21 and Gap_Lag1 at ≈ 0.16) indicates that the model heavily weighs short-term autoregressive tendencies alongside the cross-asset FX signal.

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6. Discussion of Results

6.1 What the Evidence Establishes

The empirical results provide robust evidence of a statistically significant predictive relationship from overnight FX returns to same-day equity opening gaps, which is contemporaneous at the daily frequency but exploits the temporal ordering of market closures. Across all seven experimental configurations, the HAC OLS models yielded standardized FX return betas ranging from 0.164 to 0.227, with p-values consistently below 0.025. This indicates that the observed correlation is not an idiosyncratic feature of a single firm, but rather a systematic, sector-wide phenomenon. The strength of these findings is bolstered by the use of Newey-West standard errors, which ensure that the significance levels remain valid despite the inherent autocorrelation and heteroskedasticity in financial time series.

Crucially, the evidence suggests that this information transfer is a "one-shot" event tightly bounded by the market's overnight closure. The Cross-Correlation Functions (CCFs) reveal a distinct peak at lag-0 (ranging from 0.14 for Sumitomo to 0.21 for Mitsubishi), which immediately reverses into a negative correlation at lag-1. This temporal structure is further confirmed by our Granger causality tests; with p-values for lags 1 through 5 clustered between 0.20 and 0.52, we fail to reject the null hypothesis of no causality. Note that because FX_Ret is already aligned to the same calendar day as the equity gap, the Granger test at lag 1 corresponds to the prior day's overnight FX return, effectively testing two-day-lagged information. Far from contradicting our thesis, this failure to find multi-day predictability supports the conclusion that the contemporaneous effect is confined to the immediate overnight-to-open window. The information trapped while the Tokyo Stock Exchange (TSE) is closed is released and absorbed nearly instantaneously during the opening auction, leaving no residual predictive power for subsequent sessions.

6.2 What the Evidence Does Not Establish

Despite the strong in-sample statistical association, the evidence does not support the claim that opening gap magnitudes are reliably forecastable for the purposes of high-conviction magnitude prediction. The out-of-sample (OOS) R^2 values are uniformly negative across all experiments, ranging from -0.0649 to -0.1780. In practical terms, this means that a naive zero-forecast benchmark—simply assuming no gap will occur—outperforms our regression models in a test environment. Furthermore, the Mutual Information scores remained low, peaking at approximately 0.0999, which suggests that the total statistical dependence between overnight FX returns and gap size is relatively thin once the models move beyond the training data.

Similarly, the results for unconditional directional accuracy provide a sobering perspective. While the 5711 (Mitsubishi Materials) experiment was a notable outlier with a 60.1% hit rate, the remaining tickers and baskets hovered near the 50% mark, essentially mirroring a random coin flip. These findings suggest that while a linear relationship exists on average, the signal-to-noise ratio on any given day is extremely low. It is also important to note the limitations of our sample; the data covers a relatively short period from January 2023 to the present and is restricted to three firms within a single, highly specialized sector. Consequently, the results cannot yet be generalized to other USD-sensitive industries or different currency pairs without further extensive testing across varied market regimes.

6.3 Reconciling OLS and Random Forest Evidence

A primary point of interest in our analysis is the tension between the strong in-sample HAC OLS results and the poor out-of-sample performance of the Random Forest regressor. This divergence is best understood by distinguishing between explanatory power and predictive stability. The HAC OLS models demonstrate that, within the realized sample, there is a systematic and significant association between FX returns and gap magnitudes. However, the negative OOS R^2 indicates that the Random Forest regressor likely overfits the inherent noise of the training set or struggles with the frequent regime shifts characteristic of FX-equity linkages. In the presence of

substantial volatility, the marginal explanatory signal provided by FX returns is often drowned out by idiosyncratic equity noise.

There is also a fascinating discrepancy regarding the role of FX volatility. In the linear HAC OLS specifications, FX volatility was found to be statistically insignificant, with p-values as high as 0.94. Yet, in the non-linear Random Forest models, FX volatility emerged as the second-most important feature, boasting a Mean Decrease Impurity (MDI) of approximately 0.20. This suggests that while volatility does not have a simple linear relationship with the size of the gap, it may act as a critical non-linear "gatekeeper" or conditioning variable. High overnight volatility might change how investors process directional FX returns or trigger different levels of risk aversion during the Itayose auction, creating interaction effects that OLS is mathematically unable to capture. This highlights the value of using a multi-layered econometric framework to uncover hidden structural dynamics.

6.4 Economic Mechanism

The economic logic underlying these results is centered on the fundamental cash-flow structures of the Japanese mining sector. Sumitomo Metal Mining, Mitsubishi Materials, and Mitsui Mining & Smelting all share a common financial exposure: they derive substantial revenues from precious and non-ferrous metals that are globally priced in USD, yet they maintain balance sheets and operating costs largely denominated in JPY. As noted by Bodnar and Gentry (1993), such industry characteristics make these firms naturally sensitive to exchange rate fluctuations. When the USD strengthens against the JPY overnight, the yen-denominated value of these firms' future dollar cash flows mechanically increases.

Because the TSE is closed while global FX markets continue to trade, this fundamental repricing cannot happen continuously. Instead, the information is "trapped" until the market reopens, at which point the Itayose batch auction must reconcile the previous day's close with the new overnight reality. This creates a discrete opening gap. The rapid decay of the signal—visualized by our Impulse Response Functions (IRFs) which show a significant initial shock followed by a negative reversal and a general return toward zero by steps 3–5, though individual tickers exhibit transient oscillations, is entirely consistent with the "Trading Place Hypothesis" and modern views on market efficiency. The relationship we document is not a sign of a persistent market failure, but rather a reflection of a structural information friction created by staggered trading hours. Once the TSE opens and provides a venue for equity price discovery, the FX information is absorbed, and the market returns to a state of equilibrium.

7. Quanto Options and Beyond

Implementing the signal identified in this research presents a distinct execution challenge for the cross-asset practitioner. Because the signal originates from an information friction between the currency and equity markets, any direct attempt to capture the resulting equity gap by purchasing

the underlying stock inherently exposes the trader to the very FX risk that generated the signal. A quanto option, as defined by Wystup (2006), provides a mathematically precise mechanism to isolate this effect. A quanto is a derivative whose payoff depends on a foreign underlying asset but is settled in a domestic currency at a fixed, pre-determined exchange rate, thereby neutralizing spot FX exposure at maturity.

$$\Pi_{\text{quanto}} = Q \cdot \max(S_T - K,\ 0)$$

where S_T is the stock price (JPY) at maturity,
K is the strike,
Q is the fixed USD/JPY rate

This structure is uniquely relevant to our findings because it allows a practitioner to decouple the "information-friction alpha" (the discrete repricing event at the Tokyo Stock Exchange (TSE) open) from the ongoing "FX beta" associated with holding Japanese equities.

Furthermore, our empirical results provide the necessary inputs for the sophisticated pricing of such structures. The standardized HAC OLS betas we established, which range from 0.164 to 0.227, offer a starting point for calibrating the adjustment term required in quanto pricing, which accounts for the covariance between the stock price and the exchange rate under the risk-neutral measure. By utilizing these derivatives, market participants can express a targeted view on the overnight-to-open repricing dynamics of the mining sector without commingling that view with subsequent spot FX volatility.

Future Research Directions

The contemporaneous relationship documented in this paper suggests several avenues for future work, particularly in the realms of strategy design and empirical breadth. While this study deliberately stops short of proposing a comprehensive trading strategy, our findings regarding robust directional signals and the efficacy of the lookahead guard form a natural blueprint for future implementation. Such work would benefit from integrating conditional participation filters, where trades are only initiated when classifier confidence exceeds specific thresholds.

This approach aligns with the findings of Lou, Polk, and Skouras (2019), who document that overnight and intraday returns carry distinct risk premia, suggesting that the signal we identified is a specific artifact of the market's non-trading hours. Our work extends the framework established by De Gooijer, Diks, and Gątarek (2012) by incorporating a microstructure-consistent identification strategy, specifically the Itayose auction mechanism, and a non-linear machine learning layer to better capture the complexities of cross-market information flows.

Beyond this particular metals sector, there is significant potential to explore whether similar frictions exist in other FX-sensitive industries. For example, Japanese automobile manufacturers with heavy USD exposure, or non-US oil companies whose assets are tied to the "petrodollar" while liabilities remain in local currencies, represent potential possibilities for extending this

contemporaneous analysis. Finally, the volatility puzzle revealed by our Random Forest models remains an open question. The fact that FX volatility ranks as the second-most important feature in terms of Mean Decrease Impurity despite its linear insignificance suggests a complex, non-linear conditioning role. Future research utilizing threshold or regime-switching models may uncover whether certain volatility environments amplify or dampen the transmission of overnight FX signals into equity opening gaps.

8. Conclusion

We document a statistically significant, sector-wide contemporaneous relationship between overnight USD/JPY returns and next-morning equity opening gaps. Across all seven experimental configurations—including three single-stock models and four multi-stock baskets—our models yielded standardized HAC betas ranging from 0.164 to 0.227, with p-values consistently below 0.025. These results confirm that the structural gap between the continuous 24-hour foreign exchange market and the discrete Tokyo Stock Exchange opening auction creates a measurable information friction for USD-exposed mining firms.

However, we recognize several significant limitations that temper these findings. Our sample is relatively brief, spanning from January 2023 to the present, and is restricted to three firms in a single sector and a single currency pair. Furthermore, out-of-sample magnitude prediction remains elusive; our models produced negative R^2 values in every experiment, ranging from -0.0649 to -0.1780, indicating that the relationship is subject to substantial noise. With the exception of Mitsubishi Materials (5711), which achieved a 60.1% unconditional directional accuracy, the sign of the gap remains difficult to forecast beyond a simple coin-flip in an unconditioned setting.

Despite these challenges, the convergence of evidence across multiple econometric layers provides high confidence that this is a robust, temporally bounded phenomenon rather than a statistical artifact. The presence of significant CCF lag-0 peaks, the pronounced negative reversal in impulse response functions at step-1, and the failure of Granger causality at lags 1 through 5 all point to a rapid information transfer that decays within a single trading session. Additionally, the discovery that FX volatility plays a prominent non-linear role—ranking second in Random Forest feature importance (~0.20) despite its linear insignificance—hints at a richer market dynamic than simple regression suggests.

What excites us most is the potential to transform these statistical foundations into a sophisticated trading framework. The natural next step involves acquiring intraday tick data to bridge the gap between predicting the open and executing an intraday exit. The quanto option framework offers a powerful mechanism to isolate this equity repricing signal from the underlying FX exposure, while the "volatility puzzle" presents a compelling case for dedicated investigation into regime-dependent behavior. We are eager to generalize this empirical framework to other sensitive sectors, such as Japanese automotive firms or oil majors exposed to

petrodollar dynamics, and we remain convinced that with more granular data and advanced modeling, the information friction documented here can be distilled into a practical, implementable edge.

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